

A Hilbert-Space Subcase of the Equality $\chi_{p,X} = \hbar_{p,X}$

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Status. This is a `partial_result_likely_valid` for a question in Yaroslavtsev's paper [1]. It proves the equality $\chi_{p,X} = \hbar_{p,X}$ when X is a real Hilbert space. It does not settle the general UMD Banach-space problem.

Source Question

Yaroslavtsev asks whether the equality $\chi_{p,X} = \hbar_{p,X}$, known for $X = \mathbb{R}$, holds for a general UMD Banach space X . The paper points out that a positive answer would imply a linear comparison between the UMD constant $\beta_{p,X}$ and the Hilbert-transform norm $\hbar_{p,X}$.

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Notice that if $X = \mathbb{R}$, then $\chi_{p,X} = \hbar_{p,X}$ (see Remark 3.6). Nevertheless, it remains open whether this equality holds for a general UMD Banach space X . Moreover, if this is the case, then it proves a celebrated open problem about linear dependence of the constants $\beta_{p,X}$ and $\hbar_{p,X}$, see [6, p. 48] and [15, 18, 24, 37] (so far only a square dependence is known, see (2.2)).

Proof Intuition

The only geometric input needed for Hilbert spaces is that their norm can be tested by Gaussian scalarization:

$$\mathbb{E}_\gamma |\langle \gamma, v \rangle|^p = a_p \|v\|^p.$$

Applying the scalar Hilbert transform after this random scalar projection shows that the H -valued Hilbert-transform norm is exactly the scalar one. On the other hand, both $\hbar_{p,H}$ and $\chi_{p,H}$ are at least their scalar constants because a one-dimensional Hilbert line is an isometric subspace. The source paper gives the remaining upper bound $\chi_{p,H} \leq \hbar_{p,H}$, so all constants collapse to the scalar value.

Theorem

Theorem 1. *Let H be a nonzero real Hilbert space and let $1 < p < \infty$. Then*

$$\chi_{p,H} = \hbar_{p,H} = \hbar_{p,\mathbb{R}}.$$

In particular, the equality problem from [1] has an affirmative answer for Hilbert-valued martingales.

Proof. Write $c_p = \hbar_{p,\mathbb{R}}$. We first prove $\hbar_{p,H} = c_p$.

The lower bound $\hbar_{p,H} \geq c_p$ follows by restricting to functions whose values lie in a fixed unit line $\mathbb{R}e \subset H$. For the upper bound it is enough, by density of finite-rank simple functions, to consider finite-dimensional H . Let $e_1, \dots, e_d$ be an orthonormal basis of H and write

$$f(t) = \sum_{j=1}^d f_j(t)e_j.$$

Let $\gamma_1, \dots, \gamma_d$ be independent standard real Gaussian variables, independent of t , and let

$$a_p = \mathbb{E}|\gamma_1|^p.$$

For every $v = \sum_j v_j e_j \in H$,

$$\mathbb{E}_\gamma \left| \sum_{j=1}^d \gamma_j v_j \right|^p = a_p \|v\|_H^p.$$

Using Fubini and the scalar Hilbert-transform estimate,

$$\begin{aligned} a_p \|\mathcal{H}_H f\|_{L^p(\mathbb{R};H)}^p &= \mathbb{E}_\gamma \left\| \mathcal{H}_\mathbb{R} \left(\sum_{j=1}^d \gamma_j f_j \right) \right\|_{L^p(\mathbb{R})}^p \\ &\leq c_p^p \mathbb{E}_\gamma \left\| \sum_{j=1}^d \gamma_j f_j \right\|_{L^p(\mathbb{R})}^p \\ &= a_p c_p^p \|f\|_{L^p(\mathbb{R};H)}^p. \end{aligned}$$

Thus $\hbar_{p,H} \leq c_p$, and hence $\hbar_{p,H} = c_p$.

It remains to identify $\chi_{p,H}$. Yaroslavtsev proves $\chi_{p,X} \leq \hbar_{p,X}$ for every UMD Banach space X [1, Corollary 3.5]; therefore

$$\chi_{p,H} \leq \hbar_{p,H} = c_p.$$

For the reverse inequality, again restrict the defining inequality for $\chi_{p,H}$ to integrands of the form $\Phi = \phi e$, where $e \in H$ is a unit vector and ϕ is real-valued. Any admissible constant for $\chi_{p,H}$ is then admissible for the scalar constant $\chi_{p,\mathbb{R}}$. The source paper records that $\chi_{p,\mathbb{R}} = \hbar_{p,\mathbb{R}}$ [1, p. 3 and Remark 3.6]. Hence

$$\chi_{p,H} \geq \chi_{p,\mathbb{R}} = \hbar_{p,\mathbb{R}} = c_p.$$

Combining the two inequalities gives $\chi_{p,H} = c_p = \hbar_{p,H}$. □

Verification and Novelty Notes

The proof is purely analytic; no computational check is needed. The packet uses only the source paper's inequalities and the standard Gaussian scalarization identity for Hilbert spaces.

Local cheap indexes were searched for arXiv:1812.08049, strongly orthogonal martingales, $\chi_{p,X}$, $\hbar_{p,X}$, Hilbert transform, and UMD keywords. The only close local item is an earlier attempt on arXiv:1805.03948 about broader Hilbert-transform and Burkholder-function constant problems; it does not duplicate this Hilbert-space $\chi = \hbar$ subcase. Web searches for exact $\chi_{p,X}/\hbar_{p,X}$ phrases found the source paper but no separate later answer. The Hilbert-space argument is short and may be folklore-standard, so novelty confidence is modest.

Limitations

This result does not cover non-Hilbert UMD spaces. In particular it does not prove the source paper's general equality problem, nor the linear dependence between $\beta_{p,X}$ and $h_{p,X}$ for arbitrary UMD spaces.

Human Review Recommendation

Check the finite-rank reduction in the proof of $h_{p,H} = h_{p,\mathbb{R}}$ and the scalar-line lower bound for $\chi_{p,H}$. These are the only nontrivial steps; the remaining inequalities are cited directly from the source paper.

References

- [1] I. Yaroslavl'tsev, *On strongly orthogonal martingales in UMD Banach spaces*, arXiv:1812.08049, 2018.